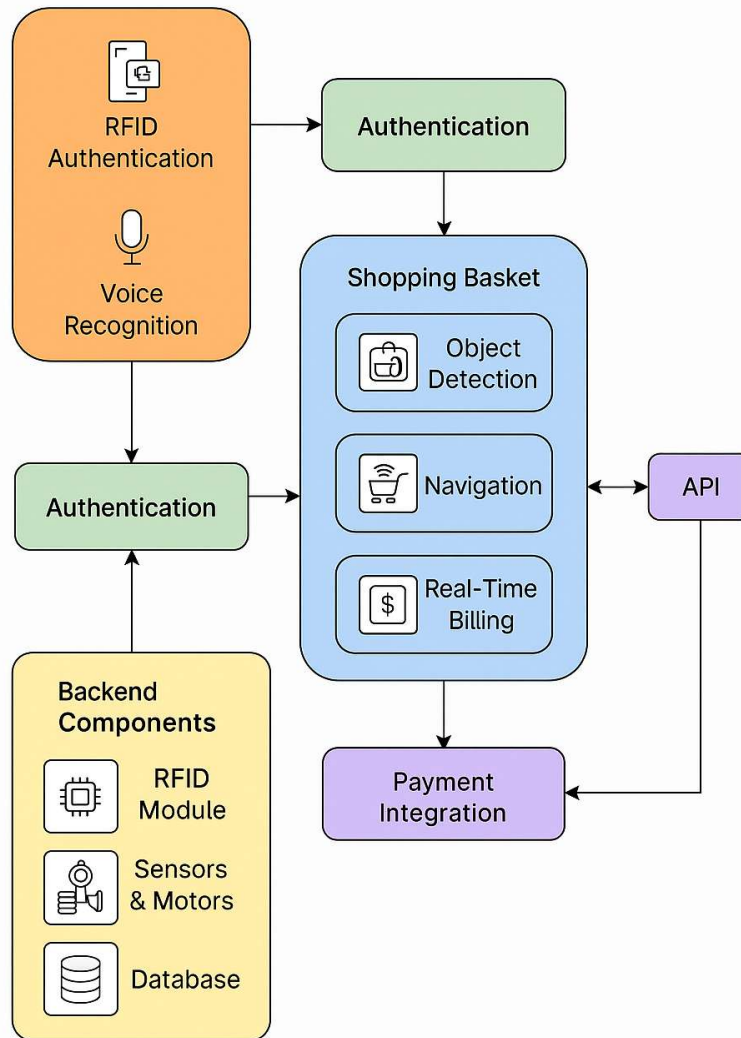




AI-Powered Touchless Shopping Basket

◆ 1. RFID Authentication

What it means:

RFID (Radio Frequency Identification) is used to identify the user by scanning their RFID tag/card.

How it works:

The basket has an **RFID reader**. When a user scans their card/tag, it retrieves a unique ID and verifies it against a database.

◆ Voice Recognition

What it means:

Capturing and interpreting user voice commands to control the basket (e.g., “Add milk”).

How it works:

Using a **microphone + speech recognition library**, the system converts spoken words to text and maps it to predefined commands like add/remove items.

■ Authentication

What it means:

The process of verifying whether a user is authorized to access and use the basket.

How it works:

Combines RFID to validate identity. If recognized, the user session is created.

◆ Shopping Basket

What it means:

This represents the smart shopping system that handles user activity after authentication.

It includes three modules:

◆ Object Detection

What it means:

AI-based detection of physical items placed in or removed from the basket.

How it works:

Using a camera + YOLO model or sensors + RFID, the system detects items being handled and identifies them.

◆ Navigation

What it means:

The basket’s ability to autonomously follow the user in the store.

How it works:

Sensors (e.g., ultrasonic, LiDAR) detect the user’s distance, and motors are controlled to follow while maintaining a safe distance.

◆ Real-Time Billing

What it means:

The live calculation of the total cost of items in the cart.

How it works:

Every time an item is added or removed (via voice, RFID, or camera), the backend updates the cart and recalculates the total using item price and quantity.

■ API

What it means:

API (Application Programming Interface) allows the **frontend** (user interface) to communicate with the **backend** (Flask server).

How it works:

Routes like /scan-rfid, /billing, /pay are exposed by Flask. The frontend calls them using JavaScript fetch().

■ Payment Integration

What it means:

Secure and contactless payment after shopping.

How it works:

Using the **Razorpay API**, the backend initiates a UPI payment and tracks whether the transaction is successful, then marks the order as “Paid.”

■ Backend Components

These are the hardware and software components powering the system:

◆ RFID Module

The hardware that reads RFID cards or tags (like MFRC522).

◆ Sensors & Motors

Hardware used to control movement (motors) and detect objects or user position (ultrasonic/LiDAR sensors).

◆ Database

Stores all data: users, products, transactions, and cart history.

Can be either:

- **MySQL** via SQLAlchemy
- **Firebase** for real-time sync and mobile support